

Exposure to bisphenol A alternatives bisphenol AF and fluorene-9-bisphenol induces gonadal injuries in male zebrafish

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ABSTRACT

Bisphenol A (BPA), present in many household products, can damage the male reproductive system. Accordingly, we summarized urine samples from 6921 human in National Health and Nutrition Examination Survey and found urinary BPA levels were inversely linked with blood testosterone in the children group. Currently, BPA replacements, such as fluorene-9-bisphenol (BHPF) and Bisphenol AF (BPAF), have been introduced to produce "BPA-free" products. Here we demonstrated that BPAF and BHPF could induce delayed gonadal migration and reduce the number of progenitors of germ cell lineage in zebrafish larvae. A close receptor analysis study reveals that BHPF and BPAF can strongly bind to androgen receptors, leading to the downregulation of meiosis-related genes and the overexpression of inflammatory markers. Furthermore, BPAF and BHPF can induce activation of the gonadal axis via negative feedback, leading to the hypersecretion of some upstream hormones and an increase in the expression of upstream hormone receptors. Our findings call for further research on the toxicological effects of BHPF and BPAF on human health and recommend that BPA replacements be investigated for anti-estrogenic action.

1. Introduction

Current male infertility accounts for 50% of all cases and affects one-sixth of families worldwide (Hirsh, 2003; Maduro and Lamb, 2002). A previous study analyzed recent trends in sperm counts from 185 studies with 42,935 semen samples collected during 1973–2011 and reported a 52.4% decrease in sperm counts (Levine et al., 2017). Environmental pollutants are one of the most serious challenges affecting human health. Concomitantly, environmental pollution has increased. This observation discovered that the components of pollutants, such as plasticizers, flame retardants, and personal care products, gradually accumulate in the human body (Griswold, 2016; M. Wang et al., 2021; W. Wang et al., 2021). There has been increasing interest in the role that environmental chemicals can influence the risk of the testis ability to produce sperm, exerting their toxic effects mainly through hormonal disruption and apoptosis (Asadi et al., 2017; Virant-Klun et al., 2022).

Bisphenol A (BPA), a very common plasticizer, is used extensively in food packaging, plastic products, and polymer materials (Wilson et al., 2007; Kubo et al., 2004; Abril et al., 2020). BPA can be detected in many environments and human biological fluids, including human urine, serum, placental tissue, amniotic fluid, and cord blood (Calafat et al., 2008; Vandenberg et al., n.d; Kang et al., 2006). Several studies have demonstrated that BPA has substantial adverse effects on the human endocrine and reproductive systems (Vandenberg et al., 2012) because of its estrogenic, antiandrogenic, and antithyroid activities (Matuszczak et al., 2019). Given the toxicity and limitations on the use of BPA (Kinch et al., 2015), many BPA analogs are used as substitutes for BPA in a range of products (Pouokam et al., 2014). Due to their similar bisphenol structure, therefore, evaluating effects of these replacements on development and reproduction is important.

Fluorene-9-bisphenol (BHPF), a typical emerging BPA alternative, is frequently used in the chemical industry as a plasticizer and combustion

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aid and in synthesizing plastics and rubber (Vandenberg et al., 2012). In mouse models, BHPF has been found to induce low uterine weight, atrophic endometria, and causes adverse pregnancy outcomes, and the corresponding doses of BPA failed to cause these adverse effects (Zhang et al., 2017a). Moreover, Bisphenol AF (BPAF) is useful for producing CF3-containing polymers with improved chemical, thermal, and mechanical properties (Li et al., 2015). BPAF has similar functions as BPA by acting as an agonist or antagonist to perturb physiological processes mediated via Estrogen receptor α (Er α) and/or Estrogen receptor β (Er β). (Matsushima et al., 2010; Zhang et al., 2017b; Yan et al., 2017; Zhao et al., 2019). Several countries have detected BPAF and BHPF in environmental, food, and human urine samples (Chen et al., 2016a; Huang et al., 2020). Exposure to these pollutants may also profoundly affect the outcomes of acute embryo exposure, liver/cardiotoxicity, and neuroendocrine toxicity, besides reproduction (W. M. Wang et al., 2021; W. Wang et al., 2021; Mi et al., 2019b; Jin et al., 2021). Unfortunately, our understanding of how exposures to these substances influence early gonadal development and male reproduction is insufficient.

In this study, we treated *vasa::RFP* transgenic zebrafish embryos (Transgenic embryos that express the red fluorescent protein (RFP) under the control of the progenitors of germ cell lineage (PGC)-specific *vasa* promoter were used to isolate pure populations of PGCs, which were purchased from Xiamen University) with BPAF and BHPF to determine the developmental and reproductive toxicities of BPAF and BHPF *in vitro*, *in vivo*, and *in silico* assays. We believe this study will extend the knowledge of environmental androgens and raise awareness of androgens BPA substitutes and the possibility of their causing adverse effects on human health.

2. Methods and materials

2.1. Demographics & statistics

The national health and nutrition examination survey (NHANES), a national, cross-sectional, population-based survey (<https://www.cdc.gov/nchs/nhanes/>), is designed to examine the nutrition and health status of Americans of all ages and races. The study population was recruited using a multistage, stratified sampling design, making it nationally representative. The analysis in the present study was comprised of demographic data, urinary environmental phenols data, and urinary sex hormone testing data from 2011 to 2016. We excluded participants with missing covariate data, women with pregnancy, and those with a BMI below 18.5. We then adjusted for BMI, occupation, and substance abuse and stratified by 12 and 18 years of age. In the end, 6921 participants fulfilled the requirements. All statistical analysis were completed before October 21, 2022 and demographic information is provided in Table S1. The National Center for Health Statistics Research Ethics Review Board approved the NHANES survey. All individuals provided written approval to participate (<https://www.cdc.gov/nchs/nhanes/biospecimens/participants.htm>).

2.1.1. Zebrafish husbandry and compounds preparation

In this work, the seven-month-old AB wild-type zebrafish (purchased from the Chinese Academy of Sciences' Institute of Aquatic Animals) and *vasa::RFP* transgenic embryos (Fan et al., 2008) were chosen. The animal experiments were conducted in accordance with the guidelines and approval of the Animal Research and Ethics Committees of Shanghai Jiao Tong University (Approval No.: SH9H-2022-A892-1). Male and female zebrafish were housed separately and fed brine shrimp twice daily in a laboratory environment with a constant temperature aeration water cycle (28 °C water temperature).

The *vasa::RFP* transgenic zygotes were cultured in an ecological habitat in a 6-well plate (100 eggs/well). They were incubated in a temperature-controlled room (28 $\pm$ 0.5 °C) with 52-lux light intensity and a 10:14 light-dark cycle. The fertilized eggs hatch after 96 h of fertilization. The 96-hour LC50 of BPAF and BHPF to zebrafish embryos

was determined using OECD Test Guideline 212. For the following test, concentrations of 1/20 and 1/10 of the 96 h LC50 were chosen. (Qiu et al., 2021). A vehicle control group (0.1% DMSO in culture medium) was prepared as appropriate. The number of surviving embryos, dead embryos, and hatchlings was tallied for four consecutive days. Embryos were observed Every 6 h in the first 48 h and twice a day in the latter 48 h under a stereoscopic microscope (Nikon, SMZ18, Japan), and removed the dead embryos as soon as observed. The *vasa::RFP* transgenic embryos were observed, and the number of progenitors of germ cells was counted using a confocal microscope.

2.1.2. Reproductive toxicity assessment

We chose 28 days as the observation point for reproductive toxicity in this experiment (Nishimura et al., 2016; Bambino and Chu, 2017). Three repetitive tanks were designed for each concentration treatment group, and put 30 fish in each concentration. Zebrafish were subjected to three levels of BPA exposure at concentrations usually found in the environment (20, 200, and 500 μ g/L). The exposure concentrations of 20 μ g/L were referenced to the measured BPA concentrations in surface water (21–85.5 μ g/L). However, because greater amounts of BPA have been documented in industrial wastewater (0.13–17200 μ g/L) and landfill leachate (8052 μ g/L), we chose 500 μ g/L as the maximum exposure dose for this study (Genheden and Ryde, 2015; Jin et al., 2021). On 14, 21, and 28 days of exposure, nine male fishes were randomly selected from each concentration treatment group every day, and the unexposed female fishes of the same batch were placed in clean water at a ratio of 1:1, separated by partitions, and placed in the mating cage. Mating behavior was completed at 9 am the next day and zebrafish embryos were collected 2 h later. The embryos were cultured in egg water (Zebrafish embryo culture solution (egg water) was standardized according to the Zebrafish Book: 137.00 mmol/L NaCl, 5.40 mmol/L KCl, 0.25 mmol/L Na2HPO4, 0.44 mmol/L K2HPO4, 1.30 mmol/L CaCl2, 1.00 mmol/L MgSO4, 4.20 mmol/L NaHCO3, and 1–2 drops of methylene blue solution.) (Westerfield M et al., 2000); then, the offspring quality was observed and recorded. After exposure, six zebrafish in each concentration group were embedded in paraffin and sectioned with a thickness of 3 μ m. Both HE staining and TUNEL staining were performed. For following investigations, 4 μ m sections were sliced and stained with hematoxylin and eosin (H&E) for histological analysis. Caspase-3 (1:200, Proteintech) was used to assess apoptosis levels in various groups. Meanwhile, the TUNEL BrightGreen Apoptosis Detection Kit (Vazyme, Nanjing, China) was used to detect apoptotic cells using terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL). To stain nuclei, 6-diamidino-2-phenylindole (DAPI) was used. A Zeiss LSM5 Live confocal microscope was used to acquire all immunofluorescence and TUNEL assay photos (Zeiss, Oberkochen, Germany). Following the treatment, Caseviewer software (3DHISTECH Ltd., Budapest, Hungary). was used to export the image.

2.1.3. Quantitative real-time polymerase chain reaction (qRT-PCR)

Six zebrafish were obtained from each concentration group. Total tissue RNA was extracted using the TRIzol (Invitrogen, Carlsbad, CA, USA), retrotranscribed into cDNA, and added to upstream and downstream primers. The SYBR Green technique was applied for quantitative fluorescence PCR. We performed real-time qPCR using the TaKaRa SYBR® Premix DimerEraser™ (Tokyo, Japan) and the Roche Light-Cycler Real-Time System to obtain cDNA using the TaKaRa Prime-Script™ RT Master Mix (Tokyo, Japan). All reactions were performed following the manufacturer's instructions. qRT-PCR was performed in triplicate by using the housekeeping gene GAPDH to standardize the mRNA levels of target genes. The primer sequence information is presented in Table S2 and purchased from Sangon Biotech (Shanghai, China).

2.1.4. Enzyme-linked immunosorbent assay (ELISA)

Five fish in each group were frozen and sedated before being

completely ground and centrifuged for 5 min at 4 °C and 7000 rpm. The supernatant (10 L) was collected for extraction and analysis. Refer to the kit for the method of hormone extraction. Typically, 390 L of Milli-Q water was added to each sample (10 L), mixed with a vortex, followed by adding 2 mL of ether, which was mixed with a mixer and centrifuged at 4 °C and 5000 rpm for 10 min to finally collect the supernatant. Following the above procedure, the supernatant was extracted once using 2 mL of ether. The second extraction supernatant was collected in a clean centrifuge tube, dried with nitrogen, and stored at − 80 °C. The reconstituted sample was analyzed for Estradiol (E2), Testosterone (T), and 11-Ketotestosterone (11-KT) according to the kit's instructions (DGBiotech™, Beijing, China).

2.1.5. Sample pretreatment and LC-MS/MS analysis of hormone-targeted metabolomics

The entire population of zebrafish was defrosted, weighed, homogenized, and centrifuged. Each defrosted fish was cut into small pieces before being placed in a centrifuge tube. Individual fish samples were homogenized at 60 Hz for 1 min and centrifuged at 12,000 rpm and 4 °C for 10 min. A hundred mg of sample was spiked with 20 µL of internal standards and 300 µL of methanol before being vortexed for 1 min and centrifuged at 14,000 g for 5 min. SPE techniques were used to pre-concentrate 700 µL of supernatant using the Oasis PRiME HLB Elution Vac Cartridge. Before SPE, 200 µL of methanol and 200 µL of water were used to condition the column. The samples were washed with 200 µL of 10% acetonitrile and n-hexane before being eluted with 40 µL of 90% acetonitrile. Before LC-MS/MS analysis, the elution was combined with 60 µL of water and stored at − 80 °C.

The LC-MS/MS system comprised a Waters Acquity UPLCTM I-Class linked to a Waters Xevo TQ-S triple quadrupole tandem mass spectrometer (Waters Corp., Milford, MA, USA). The mobile phase comprised methanol and water with 0.5 mM NH₄F (A) (B). A Waters BEH C8 column (2.1 mm, 100 mm, 1.7 µm) was coupled to an Acquity UPLCTM column in-line filter kit (0.2 µm). Using the following gradient program at a flow rate of 0.3 mL/min for 8 min at 50 °C, separation was achieved: 40% B for 0.5 min, then 55% B from 0.5 to 4 min, 75% B from 4 to 6.5 min, 95% B from 6.5 to 7.2 min, and finally 40% B from 7.2 to 8 min for column equilibrium. The volume of the injection was 10 µL. The steroid hormones were identified using the MRM mode, and Table S4 lists the selected ion transitions for each analyte and internal standard. Electrospray ionization in both positive and negative modes was used to introduce the UPLC effluent to the mass spectrometer. The following ESI-MS settings were set: capillary voltage, 2.8 kV; desolvation temperature, 550 °C; desolvation gas flow, 800 L/h; cone gas flow, 150 L/h; and Nebuliser, 7 bar. Data were gathered with a scan range of 50–500 *m/z* in centroid mode. The interscan delay was set to 20 ms, and the dwell time was set to auto. Masslynx 4.1 was used to get the data, and QuanLynx was utilized to process it (Waters Corp., Milford, MA, USA).

2.1.6. Molecular docking

The program AutoDock Vina 1.1.2 (Trott and Olson, 2010) was used to predict the binding mode of BHPF, BPA, and BPAF to the androgen receptor (AR), sex hormone binding globulin (SHBG), ERα, and Erβ proteins. The AR, SHBG, ERα, and ERβ protein crystal structures used for docking were obtained from the Protein Data Bank database (PDB ID; see Table S3). These crystal structures were created with PyMol 2.5 (Cowtan et al., 2011), including eliminating water molecules and ligands and incorporating hydrogen atoms. Then, a 22.522.522.5 (Morris et al., 2009) grid box was created to contain the complete binding site. Before molecular docking, every protein and chemical were converted to PDBQT format using AutoDock Tools 1.56. The docking settings were left at their default values, and the pose with the lowest score was extracted from the docking data for use in molecular dynamic simulation.

2.1.7. MM/GBSA calculation

The MMPBSA.py module in AMBER 18 was used for MM/GBSA (Molecular mechanics/generalized Born surface area) calculations. In this calculation (Miller et al., 2012), snapshots were taken every 100 ps from the 45–50 ns MD trajectory for binding free energy calculation, which was computed based on the equations demonstrated below:

$$\Delta G_{\text{binding}} = \Delta G_{\text{complex}} - (\Delta G_{\text{receptor}} + \Delta G_{\text{ligand}}) \quad (1)$$

$$\Delta G_{\text{binding}} = \Delta H - T\Delta S \quad (2)$$

$$\Delta H = \Delta E_{\text{vdW}} + \Delta E_{\text{ele}} + \Delta E_{\text{int}} + \Delta G_{\text{Sol}} \quad (3)$$

$$\Delta G_{\text{Sol}} = \Delta G_{\text{GB}} + \Delta G_{\text{SA}} \quad (4)$$

ΔH and $-T\Delta S$ represent the enthalpy and entropy changes upon ligand binding (Chain B binding to chain A), respectively. In Eq. (3), the enthalpy (ΔH) can be decomposed into four terms: van der Waals (ΔE_{vdW}), electrostatic interaction (ΔE_{ele}), internal energy (ΔE_{int}), and free energy of solvation (ΔG_{Sol}). ΔG_{Sol} consists of polar (ΔG_{GB}) and nonpolar solvation free energy (ΔG_{SA}). ΔG_{GB} was calculated using the GB model (Onufriev et al., 2004), and ΔG_{SA} was calculated based on the solvent-accessible surface area (SASA) estimated by the LCPO algorithm: $\Delta G_{\text{SA}} = 0.0072 \times \Delta \text{SASA}$. Furthermore, improper calculation of entropy could introduce additional errors to the $\Delta G_{\text{binding}}$ (Genheden and Ryde, 2015), so the contribution of entropy was excluded in this study.

3. Results

3.1. BPAs are correlated with reducing the testosterone level

The BPA concentration in the urine of the American population has declined yearly since 2011 (Fig. 1D), with a statistically significant difference ($P < 0.001$). In 2011 and 2012, 1064 samples stratified by gender and age demonstrated a negative connection between testosterone levels and urine bisphenol A concentration in male children [male: − 0.35, 95% confidence interval (CI): − 0.04, − 0.659], but not in adolescents and adults (Fig. 1B and C). There were no statistically significant differences between groups from 2013 to 2016.

3.2. BHPF and BPAF exposure induces the early gonadal developmental injury of vasa::RFP transgenic embryos

As displayed in Fig. 2, two typical BPA treatments caused the acute damage to development of the progenitors of germ cell lineage (PGCs) at zebrafish embryos. Using 4 dpf as the observation window, the progenitors of germ cell lineage (PGCs) development of zebrafish embryos in the exposed group was seriously disturbed, including early gonadal cell migration block and reduced number. Both compounds significantly reduced the number of PGCs colonies (white arrow in Fig. 2A) and became more toxic as the concentration of BHPF or BPAF increased (Fig. 2B). The compounds also led to an increase in mortality in the exposure group (Fig. 2C). Fig. 2E reveals the number of severe malformations in each concentration group, which is the number of embryos without normal escape reflex after stimulation, leading to an increase in malformation rate, malocclusion, pericardial edema, and spinal curvature. The poison also caused the delay of membrane rupture at 48hpf (Fig. 2D). The acute exposure experiment of embryos revealed that the two poisons could lead to the increase of mortality and malformation rate of zebrafish embryos, delayed rupture of membrane, and abnormal development of PGCs. Compared with BHPF, BPAF seemed to have greater reproductive-specific toxicity, causing early gonadal damage and more slowly ruptured membranes. However, BHPF led to higher mortality and malformation rates.

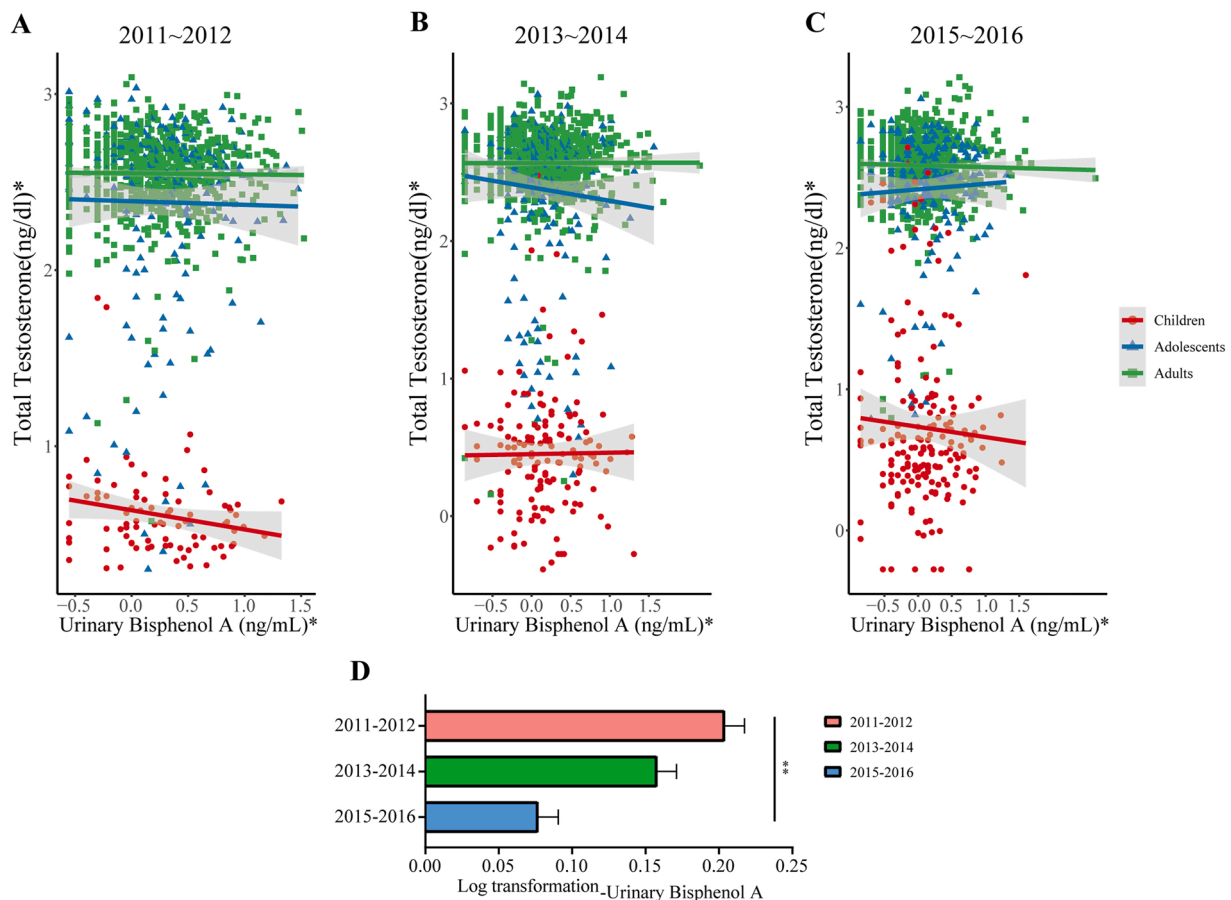


Fig. 1. Correlation between urinary BPA levels and serum testosterone (A-C). Urinary BPA levels in the population between 2011 and 2016 (D). Children and adolescents were classified according to age and sex, as defined by NHANES. Adjusted for age, race/ethnicity, urinary creatinine, total cholesterol, income, obesity, time of venipuncture, and physical activity, plus natural log-transformed BPA.

3.3. BHPF and BPAF exposure induces reproductive impairment assessment of male zebrafish

Fig. 3 A demonstrates the reproductive capacity and pathological sections of male zebrafish after the end of the exposure cycle (28 days). The spermatogenic tubules of zebrafish in the control group had complete and regular spermatogenic cells and sperm at all periods. Zebrafish in the treatment group exhibited sparse destruction of testis interstitium (yellow arrow in Fig. 3 A), decreased spermatogenic cells and spermatozoa, and vacuolated changes (green arrow in Fig. 3 A). TUNEL staining further demonstrated that apoptosis surrounded spermatogenic tubules and did not affect spermatogenic cells under low concentration of poison, suggesting the destruction of testis interstitium and blood-testosterone barrier. With the increase in concentration, fluorescence signals are diffused in spermatogenic tubules, which may be caused by poison damage to spermatogenic cells. In the mating experiment with normal female fish, we discovered that the fecundity of male zebrafish decreased significantly with the increase in drug concentration, manifested as the decrease in offspring quality. The decline in chasing behavior resulted in a decline in spawning rates (Fig. 3B). In BHPF and BPAF treatment groups, the offspring displayed an increase in the rate of abnormalities, including pericardium edema, skeletal deformity, etc. There was an increase in 96 hpf cumulative mortality. This effect on the quality of offspring increased with the time of exposure to high concentrations (500 µg/L BHPF and 500 µg/L BPAF), and no statistical difference was observed in the time of exposure to low concentrations (Fig. 3 C). Furthermore, unfertilized follicles were found after mating in the highly exposed group (Fig. S1). These findings suggest a serious decline in male fecundity, including disturbance of testis structure,

spermatogenic tubule apoptosis, and reduced offspring quality. With lower spawning rates and worse zebrafish motility than BPAF, BHPF exhibited more behaviorally disrupting effects in male zebrafish exposure trials. However, BPAF appeared to increase gonadal cell apoptosis and higher embryonic mortality, and increased offspring mortality after mating in exposed adults suggested a stronger gonadal specificity, similar to that observed in juveniles.

3.4. Gene transcription levels of adult male zebrafish after BHPF and BPAF treatment

To further examine the effects of BHPF and BPAF on adult male zebrafish, the *in vivo* expression of sex hormone receptors and key enzymes, essential inflammatory factors, and genes involved in spermatogenesis and meiosis were also studied. Both bisphenols significantly increased the expression of *fshb* (Fig. 4B), *lhb* (Fig. 4C), *era* (Fig. 4E), and *erb1* (Fig. 4F). BPAF upregulating *gnrh3* and *erb2* levels in the high concentration group (Fig. 4A and G). BHPF upregulates *erb1* levels only in the low concentration group (Fig. 4F). All doses of the toxin decreased the expression level of *cyp19a1* (Fig. 4H).

We discovered decreased spermatogenesis and elevated apoptotic signals in adult fish spermathecae, which were then analyzed for the expression of associated genes. Fig. 4I, J, and K reveal that the high concentration toxicant group elevated the expression of essential inflammatory factor genes, demonstrating a significant upregulation trend with increasing toxicant concentration. Meiosis is a crucial component of spermatogenesis. Fig. 4L-R reveal that *dmc1*, *nanog*, *cyclind1*, *aldhla2*, and *cyp26a1* were significantly downregulated in all concentration groups, and *pcna* was downregulated in BPAF (20 µg/L), BPAF (200 µg/L), and BHPF (20 µg/L) groups.

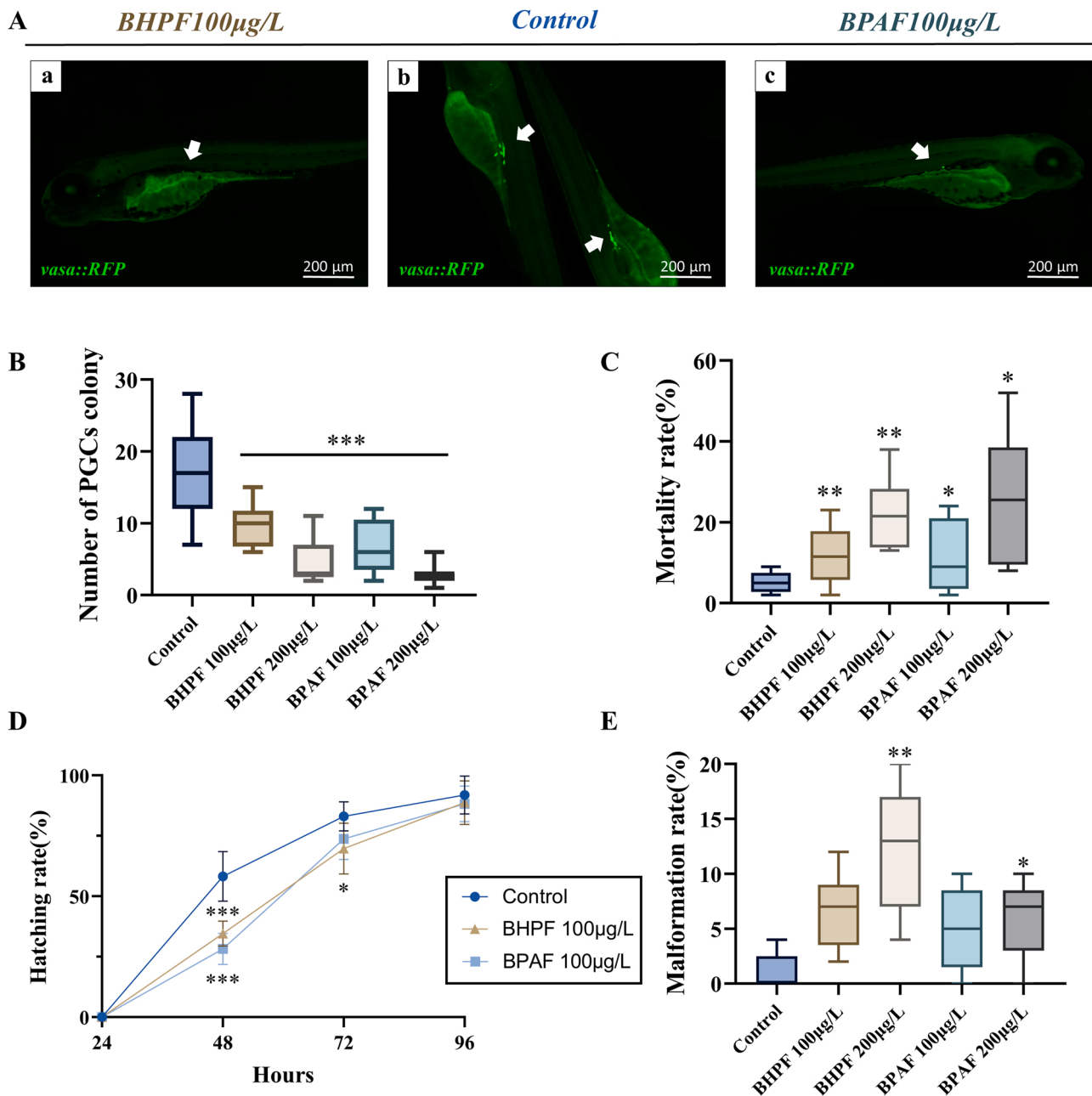


Fig. 2. Early gonadal developmental toxicity of zebrafish induced by BPAF and BHPF. Representative images of *vasa::RFP* transgenic embryos in different treatment groups at 4 dpf (A), number of PGCs colony (B), Mortality (C), Hatchability (D) and Malformation rate (E). * represents a significant difference compared with the control group (LSD test, ANOVA, *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

L), and BHPF (200 µg/L) concentration groups, and the *sycp3* gene was downregulated. In conclusion, hormone receptor and inflammatory factor-related genes exhibited a general trend of up-regulation, whereas meiosis-related genes exhibited a general tendency of down-regulation.

3.5. Sex hormones levels of adult male zebrafish after BHPF and BPAF treatment

A normal sexual hormone environment is essential for male mating behavior. Fig. 5 depicts an additional examination of hormone levels in exposed male zebrafish using metabolomics to identify the concentration of gonadal axis steroid hormones. Progesterone (Fig. 5D) and 17OH-Pregnenolone (Fig. 5F), which are upstream of hormone metabolism, were not significantly different from one another, and there was only a slight tendency toward a decrease in the high concentration of BHPF

(200 µg/L) in progesterone. However, there was a significant tendency toward an increase in the secretion of the testosterone precursors Androstenedione (Fig. 5A) and Dehydroepiandrosterone (Fig. 5B). Contrarily, increasing hazardous doses decreased the release of both testosterone (Fig. 5G) and estrogen (Fig. 5E) from the spermathecae. There was a marked rise in the release of Dihydrotestosterone (Fig. 5H), which peripheral tissues can use. Additionally, the toxin triggered the hypothalamic-pituitary-adrenal (HPA) axis, and both toxin-exposed male zebrafish had high levels of glucocorticoids in increasing amounts, indicating high levels of stress *in vivo* (Fig. 5F, I, and J). According to additional metabolomic results confirmed using ELISA, both toxicants decreased testosterone levels (Fig. 5K) in zebrafish, and BHPF decreased estradiol levels (Fig. 5M). Only at low toxicant doses (20 µg/L) testosterone 11-ketoacetate, a sex hormone precursor, exhibit an upward trend in hormone levels, which may be connected to the

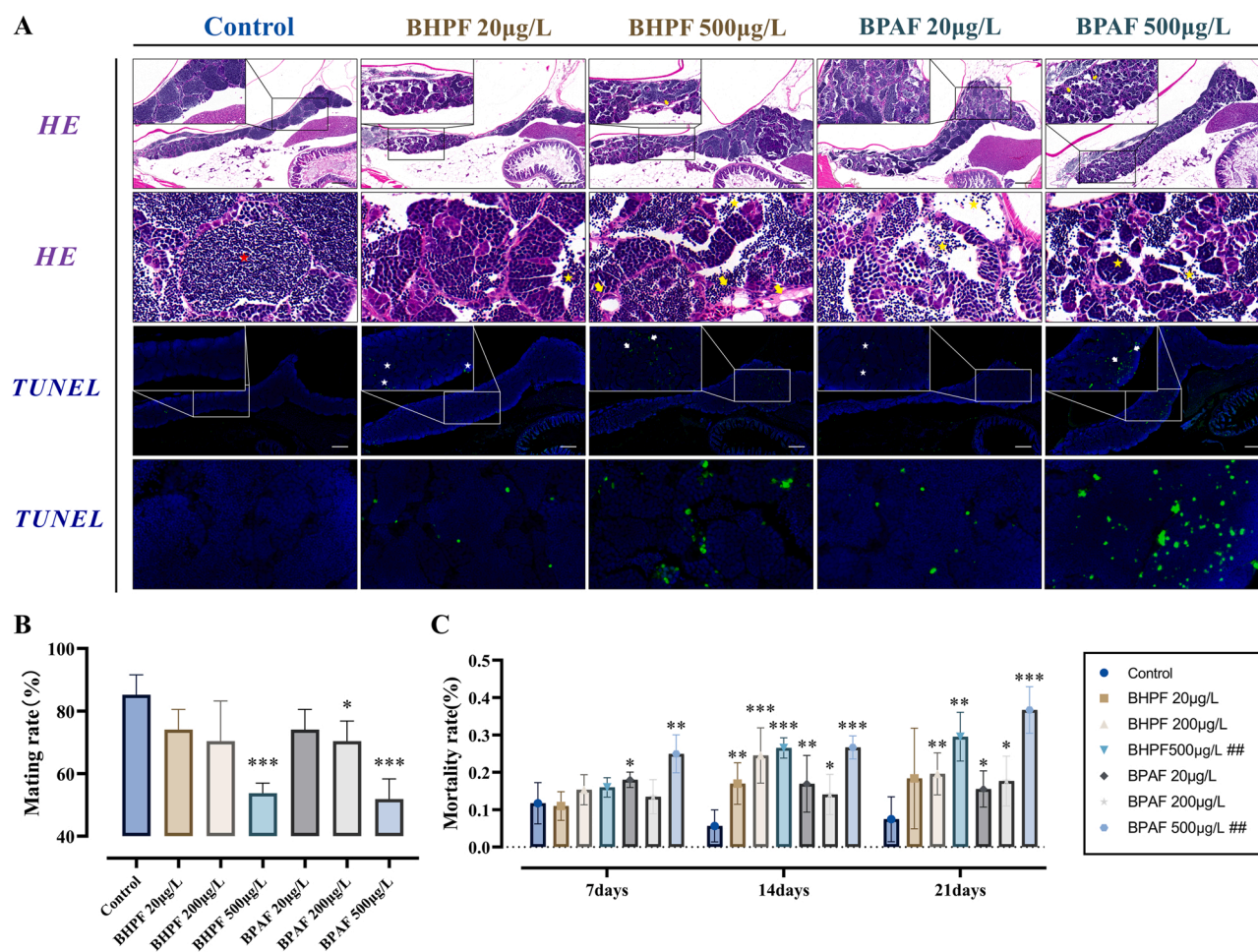


Fig. 3. BHPF and BPAF induced gonadal damage in male zebrafish. H&E staining and TUNEL staining of spermatozoa sections of toxic-exposed male fish at 28 days after toxic exposure (A); Mating rate (B) and Mortality rate (C) at 7d, 14d and 21d during the toxic exposure cycle. (LSD test, ANOVA, *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$, compared with control groups; #: $p < 0.05$; ##: $p < 0.01$; ###: $p < 0.001$, compared with same group of 7days).

compensating mechanism of sex hormones (Fig. 5L).

PCA was used to demonstrate how various groups differed metabolically. Fig. 5N reveals that PC1 and PC2 were responsible for 62.9% and 15.5%, respectively, of the overall variation. There was a distinct split between the control and other groups in the PCA score plot, suggesting that the metabolic disturbances in zebrafish varied depending on the intervention. Additionally, samples treated with two doses of BHPF were more similar to the control group than those treated with BPAF, suggesting that the intrinsic mechanisms varied, with BPAF having a larger effect on the HPA axis. Ward's linkage technique was used to calculate HCA, which was then ordered by size. Fig. 5O of HCA plot demonstrated that the five groups mentioned above could be categorized into three distinct groups: cluster I (BPAF-treated group), cluster II (control group), and cluster III (BHPF-treated group). Therefore, HCA mentioned above plot's conclusions are consistent with the findings of PCA analysis. To comprehend the biological role played by these modified steroid hormone metabolites, MetScape created the intricate pathway-based network based on these noticeably disturbed metabolites (Fig. 5P). The implicated metabolites excluded from this study were depicted by pink circles and red hexagons, respectively. Using network pharmacology technologies to identify prospective metabolic pathways, predict targets related to compounds and metabolites, and forecast targets laid the groundwork for future research on intervention mechanisms. Concerning the two drugs and ten steroid hormones, we gathered 37 common targets. Six predicted pathways, displayed in Fig. 5Q as red triangles, were most connected to the steroid hormones among all

predicted pathways. These routes were deemed essential for understanding the mechanism of action of the intervention.

3.6. BHPF and BPAF bind sex hormone receptor

Computational methods were further used to demonstrate the interaction between sex hormone receptors and BHPF and BPAF. As demonstrated above for Fig. 6A, BHPF, BPA, BPAF, and the positive reference molecule testosterone in docked binding mode with AR, BHPF formed the most interaction forces with AR. Furthermore, ASN-36 consistently forms hydrogen bonding interactions with AR, and ASN-36 may be more important for these molecules to bind to AR. Like Dihydrotestosterone, BHPF and the positive control molecule Dihydrotestosterone were attached to SHBG more strongly and primarily through hydrogen bonding and hydrophobic interactions (Fig. 6B). ASN-70 assumed its public amino acid role.

The previous binding patterns show that BHPF, BPA, and BPAF bind to the four proteins in a tolerable conformation. Nevertheless, the difference in binding strength was not discernible, and the binding stability of each system was further investigated by RMSD during MD. Fig. 6E-H depict the RMSD stability of BHPF, BPA, BPAF, and the positive reference molecules of each protein in conjunction with the proteins. The RMSD fluctuation of the molecule was less than that of the other trimeric proteins, and the RMSD value was likewise low, indicating that the molecule is more stable in AR. Among the AR proteins, the positive control molecule and BHPF float less. Among the SHBG proteins, the

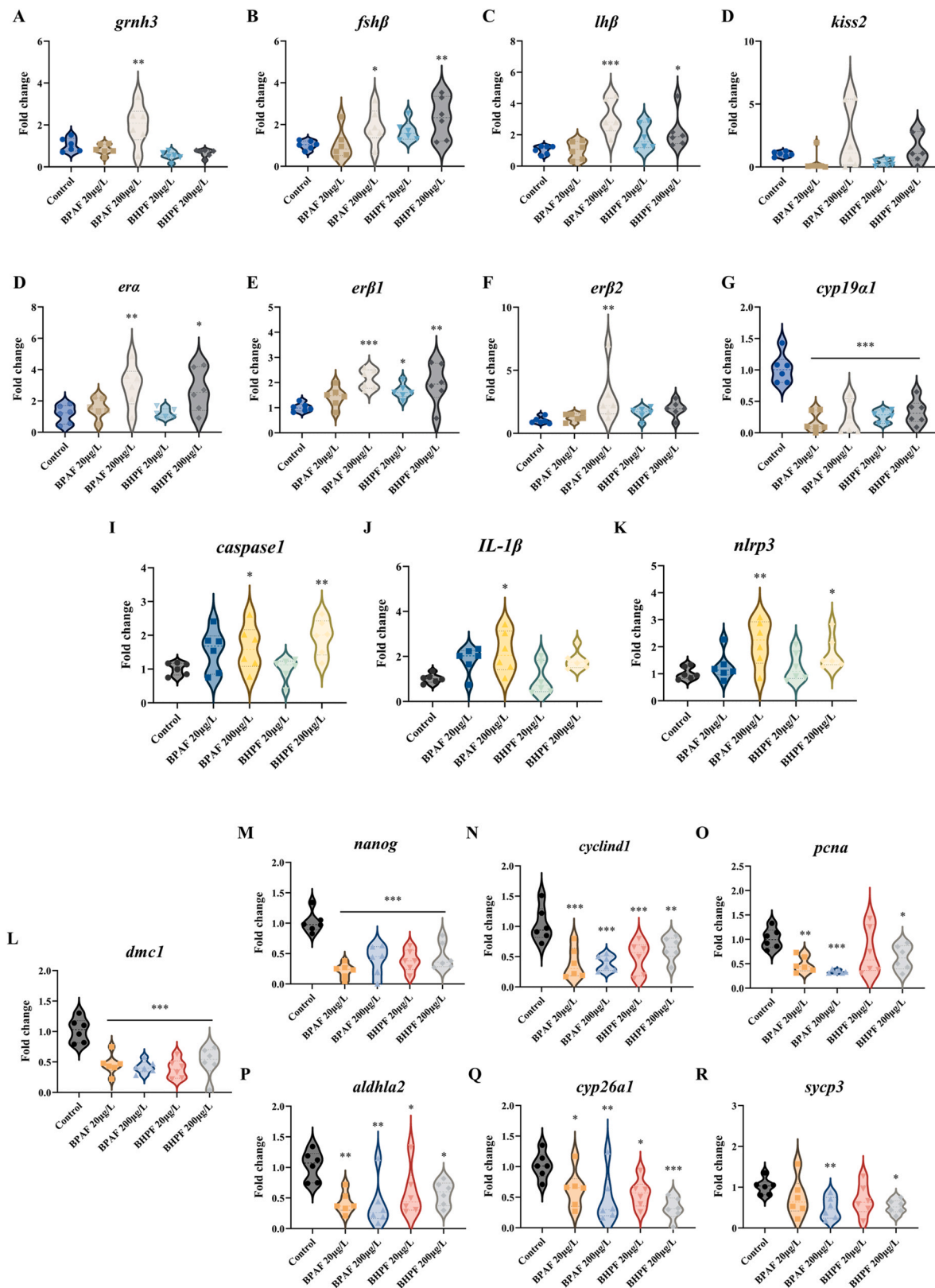


Fig. 4. The effects of BPAF and BHPF (20 and 200 µg/L) on the expression of genes encoding sex hormone receptors and key enzymes, including *grnh3* (A), *fshβ* (B), *lhβ* (C), *kiss2* (D), *era* (E), *erb1* (F), *erb2* (G) and *cyp19a1* (H) after 28-day exposure. The effects of BPAF and BHPF (20 and 200 µg/L) on the expression of genes encoding inflammatory factors, including *caspase1* (I), *IL-1β* (J), *nlrp3* (K) after 28-day exposure. The effects of BPAF and BHPF (20 and 200 µg/L) on the expression of genes encoding spermatogenesis and meiosis (in sexual reproduction), including *dmc1* (L), *nanog* (M), *cyclind1* (N), *pcna* (O), *aldhla2* (P), *cyp26a1* (Q) and *sycp3* (R) after 28-day exposure. Data are shown as the means ± standard errors of the means relative to the control group (n = 6). * represents a significant difference compared with the control group (LSD test, ANOVA, *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$).

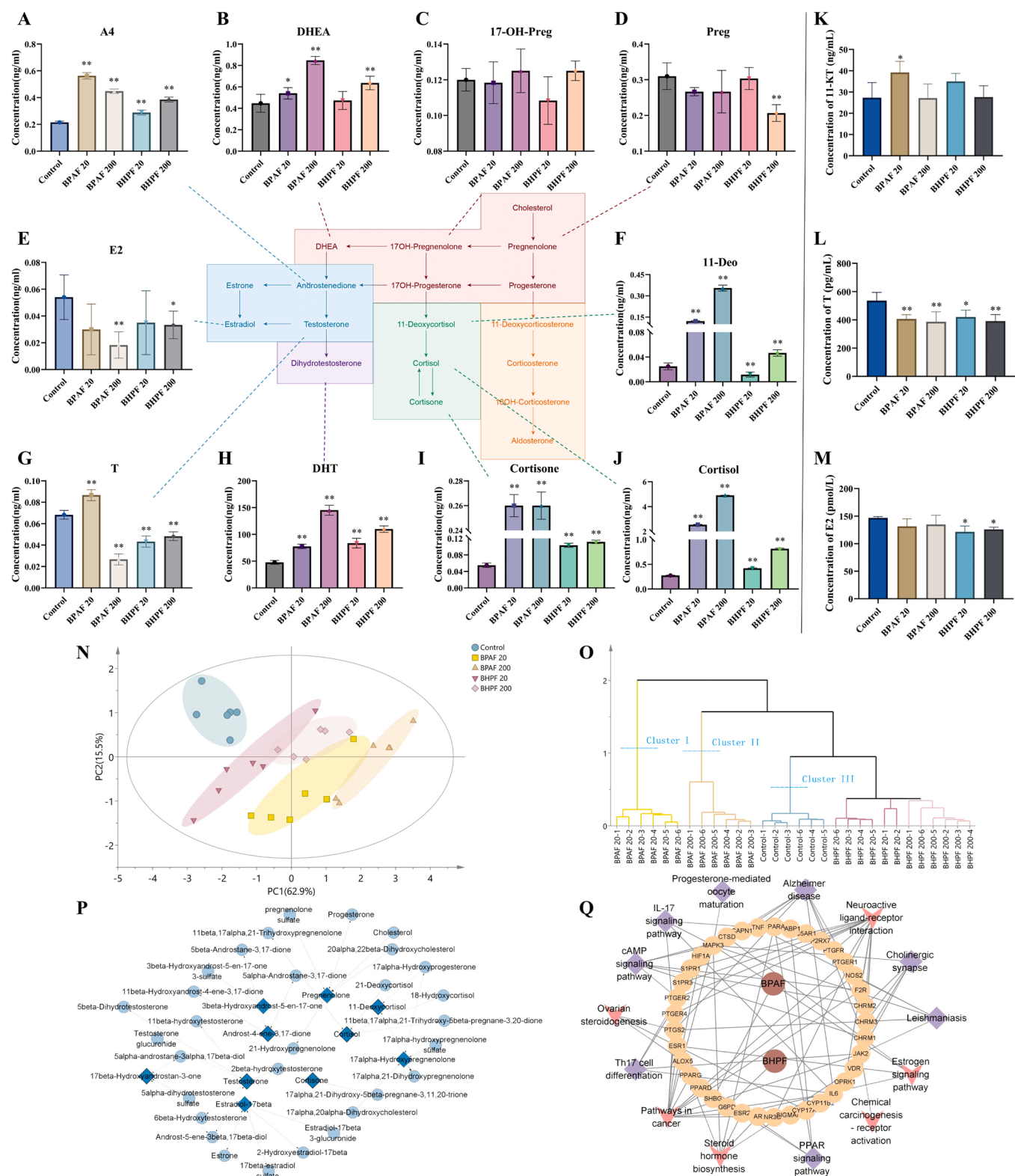


Fig. 5. Effects of BPAF and BHPF on the hormone of male zebrafish determined by metabolomics, including Androstenedione (A), Dehydroepiandrosterone (B), 17OH-Pregnenolone (C), Pregnenolone (D), Estradiol (E), 11-Deoxycortisol (F), Testosterone (G), Dihydrotestosterone (H), Cortisone (I) and Cortisol (J), expressed as mean $\pm$ SD ($n = 6$). Sex hormone disrupting effects as determined by ELISA, including concentration of 11-Ketotestosterone (K), Testosterone (L) and Estradiol (M). PCA (N), HCA (O), Pathway-based network (P) and Compound-target-pathway interaction network (Q). The red steps represent the early synthesis of sex hormones, while the green and orange are the metabolic pathways of Glucocorticoids and Mineralocorticoids, which occur in the zona reticularis, zona fasciculata and zona glomerulosa, respectively. In males, the blue steps occur mainly in the testes and include the synthesis of Testosterone and the conversion to Estrogen, actions that are performed by aromatase. Aromatase and 17 β -HSD are also present in the peripheral tissues. Finally, the purple step occurs in peripheral tissues such as the skin, prostate and epididymis, where testosterone is converted to the more potent Dihydrotestosterone. * represents a significant difference compared with control group (LSD test, ANOVA, *: $p < 0.05$; **: $p < 0.001$).

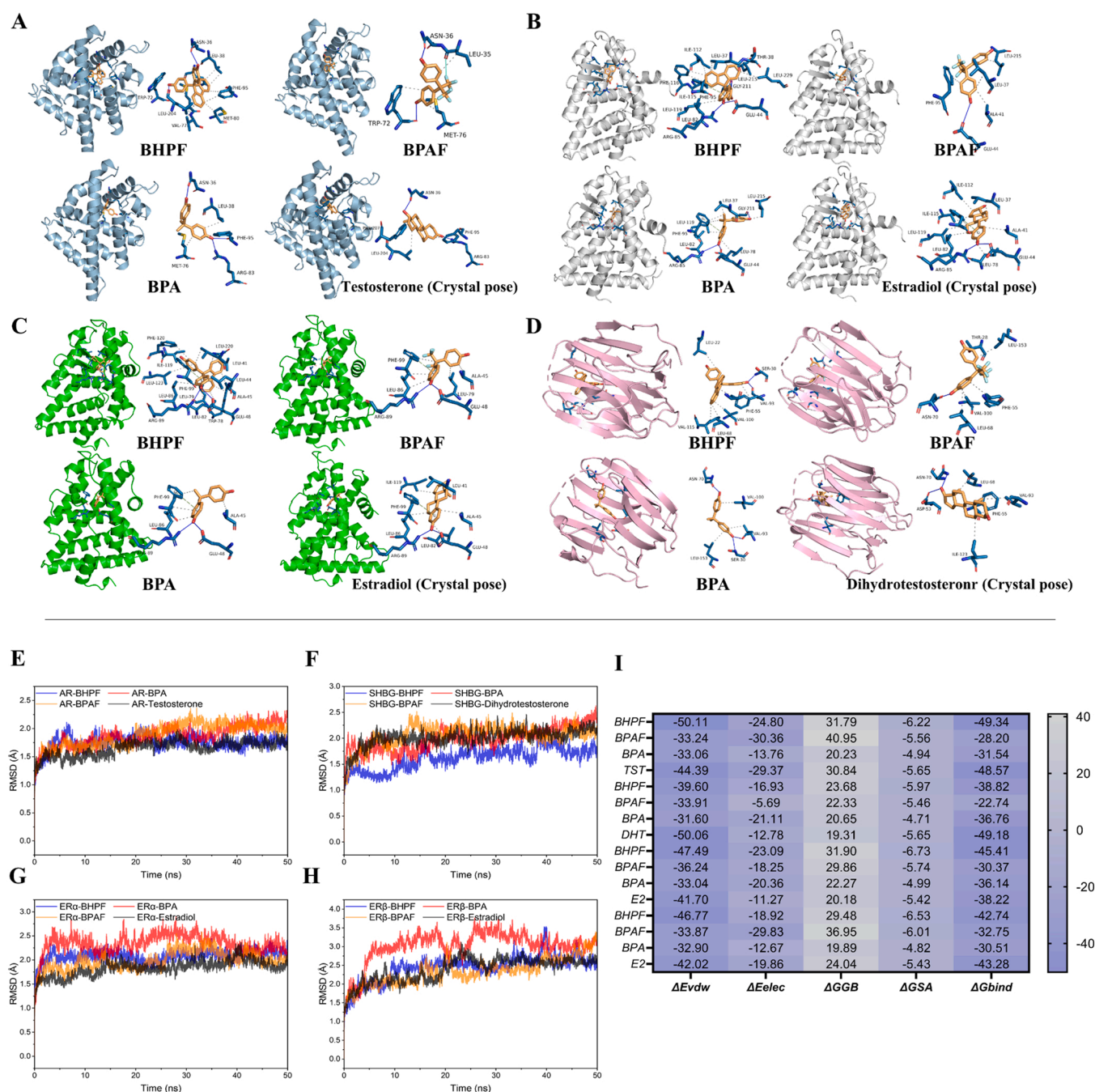


Fig. 6. The binding mode of AR (A), SHBG (B), ER α (C) and ER β (D) with each molecule. In the initial binding conformation diagram, the left is the overall view and the right is the local interaction view. In the figure, hydrogen bonds are blue lines, gray dotted lines indicate hydrophobic interaction, and green lines indicate halogen bond interaction. Change of root mean square deviation (RMSD) difference of composite with time during molecular dynamics simulation; AR complex (E); SHBG complex (F); ER α complex (G); ER β complex (H). Binding energy heatmap (I).

positive control molecule and BPAF float less. This indicates that these are the optimal binding combinations.

Based on MD trajectories, MM/GBSA is an efficient method for calculating the binding energy of small compounds to proteins, with lower values indicating better binding strength (Fig. 6I). based on the trajectory of 45–50 ns of MD simulation, we determined the binding energy for each simulated system. BHPF binds most strongly to AR, as displayed in Fig. 6F, with a binding energy of -49.34 kcal/mol, which is even slightly greater than the binding energy of testosterone to AR (-48.57 kcal/mol). It is suggested that AR may be a significant objective of the BHPF. The binding energies of BPAF and BPA to AR were -28.2 and -31.54 kcal/mol, respectively, although they were marginally

weaker for AR binding with BPAF and BPA. In the case of complexation with SHBP, the binding energies of small molecules BHPF, BPAF, and BPA were -38.82 , -22.74 , and -36.76 kcal/mol, respectively (-49.18). Among the three targets (AR, ER α , and ER β), BHPF shows a binding affinity comparable to or even stronger than the positive molecule. Furthermore, the energy breakdown reveals that the primary binding energy contribution of these compounds is the van der Waals force interaction, meaning that their binding to their particular targets is mostly driven by hydrophobic interactions.

We further examined the frequency of hydrogen bonding between small molecules and individual proteins during MD as a reflection of binding affinity and important amino acids (Table S5). The frequency of

hydrogen bonding between BHPF, BPA, BPAF, and testosterone molecules and AR proteins during MD varied, with TRP_72 hydrogen bonding at a high frequency of 0.914 for AR-BHPF, ASN_36 hydrogen bonding at a high frequency of 0.914 for AR-BPA and TRP_72 and ASN_36 can be hydrogen bonded at a high frequency of 0.914; among AR-Testosterone, ASN_36 and THR_208 can be hydrogen bonded at a high frequency of 0.914. TRP_72, ASN_36, and THR_208 are the amino acids capable of high-frequency hydrogen bonding and play a crucial role in binding these molecules.

4. Discussion

The most substantial effect of BPA on testosterone in the population was reported in 2011–2012 when an increase in BPA lowered testosterone output in male and female children under 12 years old. This result parallels an epidemiological study with a sample size of 594 individuals (Pollock et al., 2021), indicating that BPA poses a significant health risk in early development and exhibits concentration dependence. Despite the development of protective orders in numerous nations, the restrictions on BPA had little effect on the widespread usage of alternatives (Pouokam et al., 2014). Recent reports indicate that BPA alternatives, such as BPAF, Bisphenol S (BPS), Bisphenol F (BPF), and Bisphenol C (BPC), have the same hormonal action as BPA and have been identified in food and human urine samples from multiple nations (Hengstler et al., 2011; Tyl et al., 2002, 2008). Similarly, it has been demonstrated that BHPF can leach from plastic bottles into drinking water and consequently be detected in a small proportion of serum samples from human volunteers (Chen et al., 2016b). Substitutes for hazardous chemicals, in theory, should have been less dangerous than the original substance. However, some studies have indicated that exposure to BPAF, which is more similar to BPA than other analogs, affected the expression of a greater number of genes and hormone homeostasis in zebrafish embryos acting as an estrogen mimic (Qiu et al., 2021). In previous studies, treatment of mice with the same amount of BHPF as humans resulted in decreased uterine weights and reduced expression of the estrogenic response-related gene in female mice (Zhang et al., 2017a). Furthermore, in yeast two-hybrid tests and dual luciferase reporter assays, BHPF exhibited significant antiestrogenic activity. In the past two decades, estrogenic compounds have garnered significant attention as endocrine-disrupting agents (Krishnan et al., 1993; Timms et al., 2005). A vast number of natural or synthetic estrogenic compounds are classed as environmental estrogens, including BPA and BPAF. However, few potent anti-estrogenic substances have been reported, except for anti-estrogenic medications, including TAM, OHT, Raloxifene, and FULV (Routledge and Sumpter, 1997; Diamanti-Kandarakis et al., 2009; Lamb et al., 2014). Our results indicate the significant affinity of BHPF for AR and BPAF for ER, especially $ER\beta$, respectively. To investigate whether their effects differ for the same bisphenol substitutes, we compared the hormone-disrupting effects of the two toxicants on zebrafish.

In the mammalian germ cell lineage, the vasa gene has undergone relatively little change (Fan et al., 2008; Wylie, 1999). It is significant to understand the molecular control of PGC determination, proliferation, and migration in zebrafish through the vasa gene. *vasa::RFP* transgenic embryos can be a useful tool for researching the effects of toxicant exposure on the early stages of germ cell development (Yoon et al., 1997; Dumstrei et al., 2004). We confirmed that toxicants have potent sex hormone-disrupting effects on young organisms, which is consistent with our epidemiological study. In adult male zebrafish exposure experiments, the interspermatogonia matrix gradually degraded with increasing toxic concentrations. We also found abnormalities in the mating behavior of male zebrafish, including reduced chase and collision fertilization behavior. This is consistent with literature reports. The blood-testis barrier is a specialized barrier in the testis, between the interstitial blood compartment and the adluminal compartment of the seminiferous tubules, which is essential for the spermatogenic

environment to work properly (Mruk and Cheng, 2015). The barrier is formed by layers of cells from the vascular endothelium of the capillary blood vessels to the seminiferous epithelium of the seminiferous tubules. Tight junctions form between adjacent Sertoli cells, as well as between the endothelial cells (Zhou et al., 2019). Our findings reveal that toxicants cause apoptosis in spermatogonia, with low concentrations being the first to be enriched by Leydig cells with high expression of sex hormone receptors and Sertoli cells with high concentrations of the toxicant. This is consistent with previous reports of toxicant-specific damage to rodent Leydig cells (Yu et al., 2022). High concentrations of toxicant fluorescence signals were diffused in the germinal tubules, indicating an increase in a dam. In drug-treated zebrafish, qPCR revealed higher transcript levels of inflammatory factor-related genes and lower transcript levels of meiosis-related genes, supporting this assertion. According to the information mentioned above, toxicants may be enriched by the gonads due to their affinity for sex hormone receptors and result in spermatogonial damage and decreased fertility in male zebrafish.

Our *in vivo* study revealed damage to the spermatogonia's interstitium, including Leydig and Sertoli cells, the foundation for subsequent spermatocyte differentiation and the site of steroid hormone synthesis (Zhou et al., 2019), which may account for the decrease in testosterone detected by metabolomics and ELISA. However, more research is necessary to determine how toxicants affect sex hormones. Male development, maturation, and reproduction are all influenced by several mechanisms that involve the hypothalamic-pituitary-gonadal (HPG) axis (Wang et al., 2016). Overall, a reproductive function can be hampered, and clinical hypogonadism syndromes can result from any degree of congenital or acquired HPG axis dysfunction (Cunningham et al., 2012). Gonadotropin-releasing hormone (GnRH), which is released by the hypothalamus in a pulsatile fashion, stimulates luteinizing hormone (LH), gonadotropin, and insulin-like factor 3 (INSL3) to maintain the endocrine testicular function controlled by Leydig cells (Spiga et al., 2014). Whereas follicle-stimulating hormone (FSH) supports spermatogenesis, which is the process by which Sertoli cells lead the exocrine testicular function. By controlling the cholesterol hormone production pathway, GnRH, LH and FSH affect the levels of downstream hormone secretion. After receiving precise signals from upstream, the adrenocortical reticular formation creates progesterone and gradually breaks it down into precursors of testosterone. LCs in the spermatozoa are ultimately responsible for producing the effector hormone.

BPA impacts the HPG axis, which controls ovarian maturation in female goldfish (Wang et al., 2019). However, more research is still needed to determine how BHPF and BPAF affect sex hormones. Therefore, we examined changes in the expression of genes related to hormones and changes in the levels of endocrine hormones. The findings revealed that two bisphenol replacements significantly affected the zebrafish HPG axis. Notably, 200 $\mu\text{g/L}$ BPAF and BHPF treatment resulted in overexpression of nearly all genes associated with sex hormone receptors, except for *kiss2*, a crucial gene in activating neurons in zebrafish that express *GnRH3* (Qiu et al., 2016). The sex hormone-targeted metabolome demonstrated a unique separation between upstream and downstream hormone levels of the gonadal axis, and sex hormone precursors overactivity, target hormone depletion, and upstream hormone overproduction imply that BHPF and BPAF might be involved in two different damage processes. On the one hand, the poison forms a stable bond with the androgen receptor, which is then taken up by the cells that assist in spermatogenesis. The toxin activates the spermatogonial apoptotic pathway, resulting in organ damage and decreased testosterone output (Yu et al., 2022). On the other hand, the body enters a condition of androgen deficit due to the competitive binding of androgen receptors, which also activates negative feedback, causing upstream hormone production and the activation of receptor-related genes (Mhaouty-Kodja, 2018). The body enters a state of stress when the gonadal axis is overactive, and this stress causes the body to overproduce enormous levels of glucocorticoids and

hypersynthesis of steroid hormones. This is consistent with past descriptions of behavioral dysfunction in zebrafish exposed to these chemicals (Mi et al., 2019a). The body lessens the conversion of androgens to estrogens due to androgen deprivation, which also decreases hormone metabolic bypass. This results in *cyp19a1* being down-regulated, estrogen receptor genes being upregulated, and estrogen secretion being decreased. Gonadal axis hyperactivity is also a side effect of prostate cancer depot therapy. Similar hormonal changes can be seen *in vivo* when Enzalutamide and Darotamide, which are first-line therapies for prostate cancer, are used to inhibit androgen receptors and promote tumor cell death (Hussain et al., 2018)(Fizazi et al., 2019). Overall, both toxicants are capable of causing considerable reproductive damage. However, BHPF has a larger targeting impact on androgen receptors in dynamics simulation, making it more likely to be enriched by spermatozoa and induce gonadal damage. In the BHPF group, there was more gonadal damage as well as more teratogenic consequences. BPAF's impacts on the endocrine system were shown using dynamics simulation and metabolomics. BPAF caused stronger damage to the HPG and HPA axis.

This study focused solely on low and high toxicant exposures, and endocrine disruptors may exhibit non-monotonic dose-response curves, making it impossible to anticipate the clinical reaction they induce (Fagin, 2012)(Ho et al., 2017). Relevant graphs that do not precisely match the concentration patterns are also included in this report. Although earlier research suggested that these two toxins might affect estrogen receptors differently, males often experience similar hormone-disrupting effects from both toxins. This may be because estrogen levels in the body are lower in males, whose fertility relies more on androgens for homeostasis. Similarly, the study of neuroendocrine homeostasis, one of the most sophisticated regulatory systems in the human body, is still in its preliminary stages. The hormonal interactions are also multiple and complex to explain by simple feedback mechanisms. Furthermore, this article only evaluates the toxicity of a single environmental pollutant to zebrafish; humans are frequently exposed to mixtures of environmental pollutants, and their synergistic toxicity must be examined further. Currently, large-scale epidemiological data regarding the disruption of sex hormones by using BHPF and BPAF is lacking, and future research may be required. Regardless, our findings contribute to a better understanding of the effects of bisphenol on male fertility reduction and provide novel insights into the possible hazards of BHPF and BPAF to male fertility.

5. Conclusion

This study explains how BHPF and BPAF induced twofold spermatogonial injury in male zebrafish. BHPF and BPAF can target binding to gonadotropin receptors, exerting a sex hormone disrupting effect and causing hyperactivity of the gonadal axis. In addition, bisphenols can enrich the spermathecae due to their stronger receptor binding ability, damaging target organs and downstream hormone secretion, ultimately reducing fertility in adult male zebrafish. This deleterious effect on reproduction can also be extended to embryos. When these two bisphenols are exposed to vasa::RFP transgenic embryos, the early gonadal development is slowed, and fewer early gonadal cells are present. Our results suggest the safety of BPA substitutes on male reproduction and the current toxicological management of substitutes for hazardous chemicals.

CRediT authorship contribution statement

Xiangyu Meng and Xiyi Wei designed the research. Huaidong Song conducted laboratory and field work and data analysis. Tao Guo and Junjian Li assisted with experiments and graphical illustration. Shifeng Su assisted with field work and data analysis. Shangqian Wang contributed to datadiscussion. Xiangyu Meng wrote the paper with assistance from Mengjing Wang and Zengjun Wang.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2023.114634](https://doi.org/10.1016/j.ecoenv.2023.114634).

References

- Abril, C., Santos, J.L., Martín, J., Aparicio, I., Alonso, E., 2020. Occurrence, fate and environmental risk of anionic surfactants, bisphenol A, perfluorinated compounds and personal care products in sludge stabilization treatments. *Sci. Total Environ.* 711, 135048 <https://doi.org/10.1016/j.scitotenv.2019.135048>.
- Asadi, N., Bahmani, M., Kheradmand, A., Rafieian-Kopaei, M., 2017. The impact of oxidative stress on testicular function and the role of antioxidants in improving it: a review. *J. Clin. Diagn. Res.* 11, IE01–IE05. <https://doi.org/10.7860/JCDR/2017/23927.9886>.
- Bambino, K., Chu, J., 2017. Zebra Toxicol. *Environ. Health* 331–367. <https://doi.org/10.1016/bs.cdb.2016.10.007>.
- Calafat, A.M., Ye, X., Wong, L.-Y., Reidy, J.A., Needham, L.L., 2008. Exposure of the U.S. population to bisphenol A and 4-tertiary-octylphenol: 2003–2004. *Environ. Health Perspect.* 116, 39–44. <https://doi.org/10.1289/ehp.10753>.
- Chen, D., Kannan, K., Tan, H., Zheng, Z., Feng, Y.-L., Wu, Y., Widelka, M., 2016a. Bisphenol analogues other than BPA: environmental occurrence, human exposure, and toxicity—a review. *Environ. Sci. Technol.* 50, 5438–5453. <https://doi.org/10.1021/acs.est.5b05387>.
- Chen, D., Kannan, K., Tan, H., Zheng, Z., Feng, Y.-L., Wu, Y., Widelka, M., 2016b. Bisphenol analogues other than BPA: environmental occurrence, human exposure, and toxicity—a review. *Environ. Sci. Technol.* 50, 5438–5453. <https://doi.org/10.1021/acs.est.5b05387>.
- Cowtan, K., Emsley, P., Wilson, K.S., 2011. From crystal to structure with CCP4. *Acta Crystallogr. D. Biol. Crystallogr.* 67, 233–234. <https://doi.org/10.1107/S0907444911007578>.
- Cunningham, R.L., Lumia, A.R., McGinnis, M.Y., 2012. Androgen receptors, sex behavior, and aggression. *Neuroendocrinology* 96, 131–140. <https://doi.org/10.1159/000337663>.
- Diamanti-Kandaraki, E., Bourguignon, J.-P., Giudice, L.C., Hauser, R., Prins, G.S., Soto, A.M., Zoeller, R.T., Gore, A.C., 2009. Endocrine-disrupting chemicals: an Endocrine Society scientific statement. *Endocr. Rev.* 30, 293–342. <https://doi.org/10.1210/er.2009-0002>.
- Dumstrei, K., Mennecke, R., Raz, E., 2004. Signaling pathways controlling primordial germ cell migration in zebrafish. *J. Cell Sci.* 117, 4787–4795. <https://doi.org/10.1242/jcs.01362>.
- Fagin, D., 2012. Toxicology: the learning curve. *Nature* 490, 462–465. <https://doi.org/10.1038/490462a>.
- Fan, L., Moon, J., Wong, T.-T., Crodian, J., Collodi, P., 2008. Zebrafish primordial germ cell cultures derived from vasa::RFP transgenic embryos. *Stem Cells Dev.* 17, 585–597. <https://doi.org/10.1089/scd.2007.0178>.
- Fizazi, K., Shore, N., Tammela, T.L., Ulys, A., Vjaters, E., Polyakov, S., Jievals, M., Luz, M., Alekseev, B., Kuss, I., Kappeler, C., Snapir, A., Sarapohja, T., Smith, M.R., ARAMIS Investigators, 2019. Darolutamide in nonmetastatic, castration-resistant prostate cancer. *N. Engl. J. Med.* 380, 1235–1246. <https://doi.org/10.1056/NEJMoa1815671>.
- Genheden, S., Ryde, U., 2015. The MM/PBSA and MM/GBSA methods to estimate ligand-binding affinities. *Expert Opin. Drug Discov.* 10, 449–461. <https://doi.org/10.1517/17460441.2015.1032936>.

- Griswold, M.D., 2016. Spermatogenesis: the Commitment to Meiosis. *Physiol. Rev.* 96, 1–17. <https://doi.org/10.1152/physrev.00013.2015>.
- Hengstler, J.G., Foth, H., Gebel, T., Kramer, P.-J., Lilienblum, W., Schweinfurth, H., Völkel, W., Holln, K.-M., Gundert-Remy, U., 2011. Critical evaluation of key evidence on the human health hazards of exposure to bisphenol A. *Crit. Rev. Toxicol.* 41, 263–291. <https://doi.org/10.3109/10408444.2011.558487>.
- Hirsh, A., 2003. Male subfertility. *BMJ* 327, 669–672. <https://doi.org/10.1136/bmj.327.7416.669>.
- Ho, S.-M., Rao, R., To, S., Schoch, E., Tarapore, P., 2017. Bisphenol A and its analogues disrupt centrosome cycle and microtubule dynamics in prostate cancer. *Endocr. Relat. Cancer* 24, 83–96. <https://doi.org/10.1530/ERC-16-0175>.
- Huang, Z., Zhao, J.-L., Yang, Y.-Y., Jia, Y.-W., Zhang, Q.-Q., Chen, C.-E., Liu, Y.-S., Yang, B., Xie, L., Ying, G.-G., 2020. Occurrence, mass loads and risks of bisphenol analogues in the Pearl River Delta region, South China: Urban rainfall runoff as a potential source for receiving rivers. *Environ. Pollut.* 263, 114361 <https://doi.org/10.1016/j.envpol.2020.114361>.
- Hussain, M., Fizazi, K., Saad, F., Rathenborg, P., Shore, N., Ferreira, U., Ivashchenko, P., Demirhan, E., Modelska, K., Phung, D., Krivosik, A., Sternberg, C.N., 2018. Enzalutamide in men with nonmetastatic, castration-resistant prostate cancer. *N. Engl. J. Med.* 378, 2465–2474. <https://doi.org/10.1056/NEJMoa1800536>.
- Jin, M., Dang, J., Paudel, Y.N., Wang, X., Wang, B., Wang, L., Li, P., Sun, C., Liu, K., 2021. The possible hormetic effects of fluorene-9-bisphenol on regulating hypothalamic-pituitary-thyroid axis in zebrafish. *Sci. Total Environ.* 776, 145963 <https://doi.org/10.1016/j.scitotenv.2021.145963>.
- Kang, J.-H., Kondo, F., Katayama, Y., 2006. Human exposure to bisphenol A. *Toxicology* 226, 79–89. <https://doi.org/10.1016/j.tox.2006.06.009>.
- Kinch, C.D., Ibhazehiebo, K., Jeong, J.-H., Habibi, H.R., Kurrasch, D.M., 2015. Low-dose exposure to bisphenol A and replacement bisphenol S induces precocious hypothalamic neurogenesis in embryonic zebrafish. *Proc. Natl. Acad. Sci. U. S. A.* 112, 1475–1480. <https://doi.org/10.1073/pnas.1417731112>.
- Krishnan, A.V., Stathis, P., Permut, S.F., Tokes, L., Feldman, D., 1993. Bisphenol-A: an estrogenic substance is released from polycarbonate flasks during autoclaving. *Endocrinology* 132, 2279–2286. <https://doi.org/10.1210/endo.132.6.8504731>.
- Kubo, T., Maezawa, N., Osada, M., Katsumura, S., Funae, Y., Imaoka, S., 2004. Bisphenol A, an environmental endocrine-disrupting chemical, inhibits hypoxic response via degradation of hypoxia-inducible factor 1 α (HIF-1 α): structural requirement of bisphenol A for degradation of HIF-1 α . *Biochem. Biophys. Res. Commun.* 318, 1006–1011. <https://doi.org/10.1016/j.bbrc.2004.04.125>.
- Lamb, J.C., Boffetta, P., Foster, W.G., Goodman, J.E., Hentz, K.L., Rhomberg, L.R., Staveley, J., Swaen, G., Van Der Kraak, G., Williams, A.L., 2014. Critical comments on the WHO-UNEP State of the Science of Endocrine Disrupting Chemicals - 2012. *Regul. Toxicol. Pharmacol.* 69, 22–40. <https://doi.org/10.1016/j.yrtph.2014.02.002>.
- Levine, H., Jørgensen, N., Martino-Andrade, A., Mendiola, J., Weksler-Derri, D., Mindlis, I., Pinotti, R., Swan, S.H., 2017. Temporal trends in sperm count: a systematic review and meta-regression analysis. *Hum. Reprod. Update* 23, 646–659. <https://doi.org/10.1093/humupd/dmx022>.
- Li, M., Han, X., Gao, W., Chen, F., Shao, B., 2015. Bisphenol AF stimulates transcription and secretion of C-X-C chemokine ligand 12 to promote proliferation of cultured T47D breast cancer cells. *Toxicology* 338, 30–36. <https://doi.org/10.1016/j.tox.2015.09.007>.
- Maduro, M.R., Lamb, D.J., 2002. Understanding new genetics of male infertility. *J. Urol.* 168, 2197–2205. <https://doi.org/10.1097/07.ju.0000023290.61978.b2>.
- Matsushima, A., Liu, X., Okada, H., Shimohigashi, M., Shimohigashi, Y., 2010. Bisphenol AF is a full agonist for the estrogen receptor α but a highly specific antagonist for ER β . *Environ. Health Perspect.* 118, 1267–1272. <https://doi.org/10.1289/ehp.0901819>.
- Matuszczak, E., Komarowska, M.D., Debek, W., Hermanowicz, A., 2019. The impact of Bisphenol A on fertility, reproductive system, and development: a review of the literature. *Int. J. Endocrinol.* 2019, 4068717. <https://doi.org/10.1155/2019/4068717>.
- Mhaouty-Kodja, S., 2018. Role of the androgen receptor in the central nervous system. *Mol. Cell. Endocrinol.* 465, 103–112. <https://doi.org/10.1016/j.mce.2017.08.001>.
- Mi, P., Zhang, Q.-P., Li, S.-B., Liu, X.-Y., Zhang, S.-H., Li, M., Chen, D.-Y., Zhao, X., Feng, D.-F., Feng, X.-Z., 2019a. Melatonin protects embryonic development and maintains sleep/wake behaviors from the deleterious effects of fluorene-9-bisphenol in zebrafish (*Danio rerio*). *J. Pineal Res.* 66, e12530. <https://doi.org/10.1111/jpi.12530>.
- Mi, P., Zhang, Q.-P., Zhang, S.-H., Wang, C., Zhang, S.-Z., Fang, Y.-C., Gao, J.-Z., Feng, D.-F., Chen, D.-Y., Feng, X.-Z., 2019b. The effects of fluorene-9-bisphenol on female zebrafish (*Danio rerio*) reproductive and exploratory behaviors. *Chemosphere* 228, 398–411. <https://doi.org/10.1016/j.chemosphere.2019.04.170>.
- Miller, B.R., McGee, T.D., Swails, J.M., Homeyer, N., Gohlke, H., Roitberg, A.E., 2012. MMPBSA.py: an efficient program for end-state free energy calculations. *J. Chem. Theory Comput.* 8, 3314–3321. <https://doi.org/10.1021/ct300418h>.
- Morris, G.M., Huey, R., Lindstrom, W., Sanner, M.F., Belew, R.K., Goodsell, D.S., Olson, A.J., 2009. AutoDock4 and AutoDockTools4: automated docking with selective receptor flexibility. *J. Comput. Chem.* 30, 2785–2791. <https://doi.org/10.1002/jcc.21256>.
- Mruk, D.D., Cheng, C.Y., 2015. The mammalian blood-testis barrier: its biology and regulation. *Endocr. Rev.* 36, 564–591. <https://doi.org/10.1210/er.2014-1101>.
- Nishimura, Y., Inoue, A., Sasagawa, S., Koiki, J., Kawaguchi, K., Kawase, R., Maruyama, T., Kim, S., Tanaka, T., 2016. Using zebrafish in systems toxicology for developmental toxicity testing. *Congenit. Anom. (Kyoto)* 56, 18–27. <https://doi.org/10.1111/cga.12142>.
- Onufriev, A., Bashford, D., Case, D.A., 2004. Exploring protein native states and large-scale conformational changes with a modified generalized born model. *Proteins* 55, 383–394. <https://doi.org/10.1002/prot.20033>.
- Pollock, T., Arbuckle, T.E., Guth, M., Bouchard, M.F., St-Amand, A., 2021. Associations among urinary triclosan and bisphenol A concentrations and serum sex steroid hormone measures in the Canadian and U.S. Populations. *Environ. Int.* 146, 106229. <https://doi.org/10.1016/j.envint.2020.106229>.
- Poukam, G.B., Ajaeji, G.C., Mantovani, A., Orisakwe, O.E., Frazzoli, C., 2014. Use of Bisphenol A-containing baby bottles in Cameroon and Nigeria and possible risk management and mitigation measures: community as milestone for prevention. *Sci. Total Environ.* 481, 296–302. <https://doi.org/10.1016/j.scitotenv.2014.02.026>.
- Qiu, W., Zhao, Y., Yang, M., Farajzadeh, M., Pan, C., Wayne, N.L., 2016. Actions of Bisphenol A and Bisphenol S on the reproductive neuroendocrine system during early development in zebrafish. *Endocrinology* 157, 636–647. <https://doi.org/10.1210/en.2015-1785>.
- Qiu, W., Liu, S., Chen, H., Luo, S., Xiong, Y., Wang, X., Xu, B., Zheng, C., Wang, K.-J., 2021. The comparative toxicities of BPA, BPB, BPS, BPF, and BPAF on the reproductive neuroendocrine system of zebrafish embryos and its mechanisms. *J. Hazard. Mater.* 406, 124303. <https://doi.org/10.1016/j.jhazmat.2020.124303>.
- Routledge, E.J., Sumpter, J.P., 1997. Structural features of alkylphenolic chemicals associated with estrogenic activity. *J. Biol. Chem.* 272, 3280–3288. <https://doi.org/10.1074/jbc.272.6.3280>.
- Spiga, F., Walker, J.J., Terry, J.R., Lightman, S.L., 2014. HPA Axis-Rhythms. In: *Comprehensive Physiology*. Wiley, pp. 1273–1298. <https://doi.org/10.1002/cphy.c140003>.
- Timms, B.G., Howdeshell, K.L., Barton, L., Bradley, S., Richter, C.A., vom Saal, F.S., 2005. Estrogenic chemicals in plastic and oral contraceptives disrupt development of the fetal mouse prostate and urethra. *Proc. Natl. Acad. Sci. U. S. A.* 102, 7014–7019. <https://doi.org/10.1073/pnas.0502544102>.
- Trott, O., Olson, A.J., 2010. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* 31, 455–461. <https://doi.org/10.1002/jcc.21334>.
- Tyl, R.W., Myers, C.B., Marr, M.C., Thomas, B.F., Keimowitz, A.R., Brine, D.R., Veselica, M.M., Fail, P.A., Chang, T.Y., Seely, J.C., Joiner, R.L., Butala, J.H., Dimond, S.S., Cagen, S.Z., Shiotsuka, R.N., Stropp, G.D., Waechter, J.M., 2002. Three-generation reproductive toxicity study of dietary bisphenol A in CD Sprague-Dawley rats. *Toxicol. Sci.* 68, 121–146. <https://doi.org/10.1093/toxsci/68.1.121>.
- Tyl, R.W., Myers, C.B., Marr, M.C., Sloan, C.S., Castillo, N.P., Veselica, M.M., Seely, J.C., Dimond, S.S., Van Miller, J.P., Shiotsuka, R.N., Beyer, D., Hentges, S.G., Waechter, J.M., 2008. Two-generation reproductive toxicity study of dietary bisphenol A in CD-1 (Swiss) mice. *Toxicol. Sci.* 104, 362–384. <https://doi.org/10.1093/toxsci/kfn084>.
- Vandenberg, L.N., Hauser, R., Marcus, M., Olea, N., Welshons, W.V., n.d. Human exposure to bisphenol A (BPA). *Reprod. Toxicol.* 24, 139–77. <https://doi.org/10.1016/j.reprotox.2007.07.010>.
- Vandenberg, L.N., Colborn, T., Hayes, T.B., Heindel, J.J., Jacobs, D.R., Lee, D.-H., Shioda, T., Soto, A.M., vom Saal, F.S., Welshons, W.V., Zoeller, R.T., Myers, J.P., 2012. Hormones and endocrine-disrupting chemicals: low-dose effects and nonmonotonic dose responses. *Endocr. Rev.* 33, 378–455. <https://doi.org/10.1210/er.2011-1050>.
- Virant-Klun, I., Imamovic-Kumalic, S., Pinter, B., 2022. From Oxidative Stress to Male Infertility: Review of the Associations of Endocrine-Disrupting Chemicals (Bisphenols, Phthalates, and Parabens) with Human Semen Quality. *Antioxid. (Basel, Switz.)* 11. <https://doi.org/10.3390/antiox11081617>.
- Wang, M., Li, Q., Hou, M., Chan, L.L.Y., Liu, M., Ter, S.K., Dong, T., Xia, Y., Chotirmall, S.H., Fang, M., 2021. Inactivation of common airborne antigens by perfluoroalkyl chemicals modulates early life allergic asthma. *Proc. Natl. Acad. Sci. U. S. A.* 118. <https://doi.org/10.1073/pnas.2011957118>.
- Wang, Q., Yang, H., Yang, M., Yu, Y., Yan, M., Zhou, L., Liu, X., Xiao, S., Yang, Y., Wang, Y., Zheng, L., Zhao, H., Li, Y., 2019. Toxic effects of bisphenol A on goldfish gonad development and the possible pathway of BPA disturbance in female and male fish reproduction. *Chemosphere* 221, 235–245. <https://doi.org/10.1016/j.chemosphere.2019.01.033>.
- Wang, R., Chon, H., Lee, S., Cheng, Z., Hong, S.H., Yoon, Y.H., Choo, J., 2016. Highly Sensitive Detection of Hormone Estradiol E2 Using Surface-Enhanced Raman Scattering Based Immunoassays for the Clinical Diagnosis of Precocious Puberty. *ACS Appl. Mater. Interfaces* 8, 10665–10672. <https://doi.org/10.1021/acsami.5b10996>.
- Wang, W., Hu, Y.-F., Pang, M., Chang, N., Yu, C., Li, Q., Xiong, J.-W., Peng, Y., Zhang, R., 2021. BMP and Notch Signaling Pathways differentially regulate Cardiomyocyte Proliferation during Ventricle Regeneration. *Int. J. Biol. Sci.* 17, 2157–2166. <https://doi.org/10.7150/ijbs.59648>.
- Wilson, N.K., Chuang, J.C., Morgan, M.K., Lordo, R.A., Sheldon, L.S., 2007. An observational study of the potential exposures of preschool children to pentachlorophenol, bisphenol-A, and nonylphenol at home and daycare. *Environ. Res.* 103, 9–20. <https://doi.org/10.1016/j.envres.2006.04.006>.
- Wylie, C., 1999. Germ cells. *Cell* 96, 165–174. [https://doi.org/10.1016/s0092-8674\(00\)80557-7](https://doi.org/10.1016/s0092-8674(00)80557-7).
- Yan, Z., Liu, Y., Yan, K., Wu, S., Han, Z., Guo, R., Chen, M., Yang, Q., Zhang, S., Chen, J., 2017. Bisphenol analogues in surface water and sediment from the shallow Chinese freshwater lakes: Occurrence, distribution, source apportionment, and ecological and human health risk. *Chemosphere* 184, 318–328. <https://doi.org/10.1016/j.chemosphere.2017.06.010>.
- Yoon, C., Kawakami, K., Hopkins, N., 1997. Zebrafish vasa homologue RNA is localized to the cleavage planes of 2- and 4-cell-stage embryos and is expressed in the primordial germ cells. *Development* 124, 3157–3165. <https://doi.org/10.1242/dev.124.16.3157>.

- Yu, Y., Xin, X., Ma, F., Li, X., Wang, Y., Zhu, Q., Chen, H., Li, H., Ge, R.-S., 2022. Bisphenol AF blocks Leydig cell regeneration from stem cells in male rats. *Environ. Pollut.* 298, 118825. <https://doi.org/10.1016/j.envpol.2022.118825>.
- Zhang, Z., Hu, Y., Guo, J., Yu, T., Sun, L., Xiao, X., Zhu, D., Nakanishi, T., Hiromori, Y., Li, Junyu, Fan, X., Wan, Y., Cheng, S., Li, Jun, Guo, X., Hu, J., 2017a. *Nat. Commun.* 14585. <https://doi.org/10.1038/ncomms14585>.
- Zhang, Z., Hu, Y., Guo, J., Yu, T., Sun, L., Xiao, X., Zhu, D., Nakanishi, T., Hiromori, Y., Li, Junyu, Fan, X., Wan, Y., Cheng, S., Li, Jun, Guo, X., Hu, J., 2017b. Fluorene-9-bisphenol is anti-oestrogenic and may cause adverse pregnancy outcomes in mice. *Nat. Commun.* 14585. <https://doi.org/10.1038/ncomms14585>.
- Zhao, X., Qiu, W., Zheng, Y., Xiong, J., Gao, C., Hu, S., 2019. Occurrence, distribution, bioaccumulation, and ecological risk of bisphenol analogues, parabens and their metabolites in the Pearl River Estuary, South China. *Ecotoxicol. Environ. Saf.* 180, 43–52. <https://doi.org/10.1016/j.ecoenv.2019.04.083>.
- Zhou, R., Wu, J., Liu, B., Jiang, Y., Chen, W., Li, J., He, Q., He, Z., 2019. The roles and mechanisms of Leydig cells and myoid cells in regulating spermatogenesis. *Cell. Mol. Life Sci.* 76, 2681–2695. <https://doi.org/10.1007/s00018-019-03101-9>.